Exam

Exercise 1

Based on your reading of the article "Deep Unfolding of a Proximal Interior Point Method for Image Restoration" by C. Bertocchi *et al.*, answer the following questions :

1. Consider an inverse problem related to model (1). Give two difficulties that one faces when trying to estimate $\bar{x}$ from y with a variational approach, *i.e.* by formulating and solving a minimization problem.
2. (a) Assuming that $(\forall i \in \{1, \dots, p\})\ -c_i \in \Gamma_0(\mathbb{R}^n)$, check that the logarithmic barrier defined in (9) is convex.
   (b) Is $\mathcal{B}$ differentiable on $\mathbb{R}^n$? Is it gradient-Lipschitz ? Could the gradient descent algorithm be used to solve (8) ?
3. (a) Explain why we need to compute the differential of the proximity operator of the barrier in order to train iRestNet.
   (b) Consider the following feasible domain $\mathcal{C} = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$ and its associated logarithmic barrier $\mathcal{B}$. Let $\varphi : (x, \gamma) \mapsto \operatorname{prox}_{\gamma\mathcal{B}}(x)$. For every $x \in \mathbb{R}$ and $\gamma > 0$ give the expression for $\varphi(x, \gamma)$, $\frac{\partial \varphi}{\partial x}(x, \gamma)$ and $\frac{\partial \varphi}{\partial \gamma}(x, \gamma)$.
4. The authors claim that 'every standard activation function can be derived from a proximity operator.' Prove that the latter is true, for instance, for the Rectified Linear Unit (ReLU) activation function defined below

$$\mathrm{ReLU} : \mathbb{R} \to \mathbb{R} : z \mapsto \begin{cases} z & \text{if } z > 0 \\ 0 & \text{if } z \leq 0 \end{cases}. \tag{1}$$

5. 'One critical issue concerning neural networks is to guarantee that their performance remain acceptable when the input is perturbed.'
   Consider a neural network modelled as an operator $T : \mathbb{R}^n \to \mathbb{R}^n$. Assume that T is α-averaged and that the input $x \in \mathbb{R}^n$ is subject to a perturbation $\delta_x \in \mathbb{R}^n$. Let $\Delta T(x) = T(x + \delta_x) - T(x)$.
   (a) Derive an upperbound for the norm of the output perturbation $\|\Delta T(x)\|$.
   (b) What is the limit of $\Delta T(x)$ as α tends to 0.
6. Consider the neural network $T = R_{K-1} \circ (W_{K-1} \cdot + b_{K-1}) \circ \ldots \circ R_0 \circ (W_0 \cdot + b_0)$ where $(W_k, b_k, R_k)_{0 \leq k \leq K-1}$ are defined in (61). Let $\gamma > 0$ and $\lambda > 0$, and assume that $(\forall k \in \{0, \dots, K-1\})\ \gamma_k = \gamma$ and $\lambda_k = \lambda$.

(a) Show that $(\forall k \in \{1, \dots, K-1\})\ \theta_k = 2^{k-1}\|W_0\|^k$, where θ_k is defined in Proposition 4.

(b) Show that, if (66) is satisfied, then $\|W_0\| \leq 1$.

(c) Fix $\lambda > 0$, show that $\|W_0\| \leq 1$ if and only if γ satisfies a given condition. Note that this condition is the same as the one required regarding the stepsize for the convergence of the projected gradient algorithm.

(d) Prove that, if the condition derived in the previous question is satisfied, then the operator T is nonexpansive.

7. In the numerical experiments, what is the training loss used to train iRestNet ? Is the proposed neural network trained to output a solution to (79) ? to the associated unconstrained subproblem (8) ?

Exercise 2

In binary classification, a useful loss function is the *logistic loss*, defined as :

$$(\forall x \in \mathbb{R}^n) \quad f(x) = \sum_{i=1}^{m} \log(1 + \exp(-y_i x^\top \theta_i))$$

with $n \geq 1$ the number of parameters of the classifier, $m \geq 1$ the number of feature vectors, and for every $i \in \{1, \dots, m\}$, y_i equals -1 or $+1$ is the label associated to the feature vector $\theta_i \in \mathbb{R}^n$.

1. Find $L \in \mathbb{R}^{m \times n}$ such that f reads :

$$(\forall x \in \mathbb{R}^n) \quad f(x) = \sum_{i=1}^{m} \ell([Lx]_i)$$

with $\ell : x \mapsto \log(1 + \exp(x))$ and, for every $i \in \{1, \dots, m\}$, for every $x \in \mathbb{R}^n$, $[Lx]_i \in \mathbb{R}$ denotes the i-th entry of vector $Lx \in \mathbb{R}^m$.

2. Prove that $\ell \in \Gamma_0(\mathbb{R})$. Deduce that $f \in \Gamma_0(\mathbb{R}^n)$.
3. Discuss the existence and unicity of the minimizers of f (if needed, give conditions on matrix L).
4. Prove that ℓ is Lipschitz differentiable with constant $\beta = \frac{1}{4}$.
5. Deduce that f is Lipschitz differentiable on $\mathbb{R}^n$. Precise the value of the associated constant.

We consider the penalized logistic loss, defined as :

$$(\forall x \in \mathbb{R}^n) \quad h(x) = f(x) + g(x)$$

with

$$(\forall x \in \mathbb{R}^n) \quad g(x) = \lambda \|x\|^2,$$

with $\lambda > 0$.

1. Prove that $h \in \Gamma_0(\mathbb{R}^n)$.

2. Discuss the existence and unicity of the minimizers of h.
3. Show that h is Lipschitz differentiable on $\mathbb{R}^n$. Precise the value of the associated constant.
4. Deduce a gradient descent algorithm to minimize h. Give the conditions that ensure the convergence of the iterates to a minimizer of h.

A possible strategy to accelerate this algorithm is to make use of a majorization-minimization (MM) strategy. Let $u \in \mathbb{R}$. We have the following majorization inequality :

$$(\forall v \in \mathbb{R}) \quad \ell(v) \leq \ell(u) + \nabla\ell(u)(u-v) + \frac{\omega(u)}{2}(u-v)^2$$

with $\omega(u) = \frac{1}{u}\left(\frac{1}{1+\exp(-u)} - \frac{1}{2}\right)$ if $u \neq 0$, $\omega(0) = \frac{1}{4}$.

1. Let $z \in \mathbb{R}^n$. Deduce from the above inequality the existence of a symmetric definie positive matrix $A(z) \in \mathbb{R}^{n\times n}$ such that

$$(\forall x \in \mathbb{R}^n) \quad h(x) \leq h(z) + \nabla h(z)^\top (x-z) + \frac{1}{2}(x-z)^\top A(z)(x-z).$$

2. Deduce a MM quadratic algorithm to minimize function h. Give the conditions that ensure the convergence of the iterates to a minimizer of h.
3. Give the name and principle of a possible strategy, for reducing the complexity of this algorithm.

Now, assume that we take $g = \lambda||x||_1$ instead, in the expression of the penalized loss.

1. Give the names of two possible algorithms, seen in the course, to solve the problem of minimizing h. Explain briefly their principle, their advantages and drawbacks in the context of large scale optimization.